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Problem Statement ID: SIH26073

Problem Statement Title: AI/ML-Based Intelligent Anomaly Detection for Automatic Weather Stations (AWS)

Theme: Disaster Management

PS Category: Software

Team ID:

Team Name: **SkySentinels**

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PROPOSED SOLUTION

SkyGuard AI

Real-time explainable anomaly detection for temperature, pressure & humidity sensors on AWS networks

- Temperature
- Pressure
- Humidity

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IDEA TITLE

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SkyGuard AI: *Intelligent Real-Time Anomaly Detection for Temperature, Pressure & Humidity Sensors in Automatic Weather Stations*

PROPOSED SOLUTION

- A lightweight, explainable AI/ML engine that continuously screens AWS temperature, pressure & humidity streams and flags faulty or inconsistent readings before they reach forecasters.
- Learns each station's normal diurnal & seasonal rhythm, then checks new readings against it and against neighbouring stations for spatial consistency.
- Classifies flagged readings as spikes, frozen values, drift, or communication errors, and generates a confidence score with SHAP-based reasoning.
- Runs centrally on streaming data and, in a lightweight form, directly on ESP32-class edge hardware at the station.

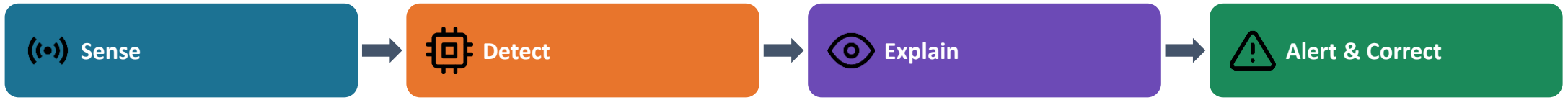
WHAT MAKES IT DIFFERENT

Multivariate, not single-sensor
Cross-checks temperature, pressure & humidity together, so a genuine storm isn't mistaken for a faulty sensor.

Explainable by design
Every alert ships with a SHAP/LIME reason — not just a red flag — so operators trust and act on it.

Edge + network aware
Same model scales from one ESP32-powered station to a nationwide AWS network.

Self-healing suggestions
Where possible, proposes a corrected/imputed value instead of just discarding the reading.



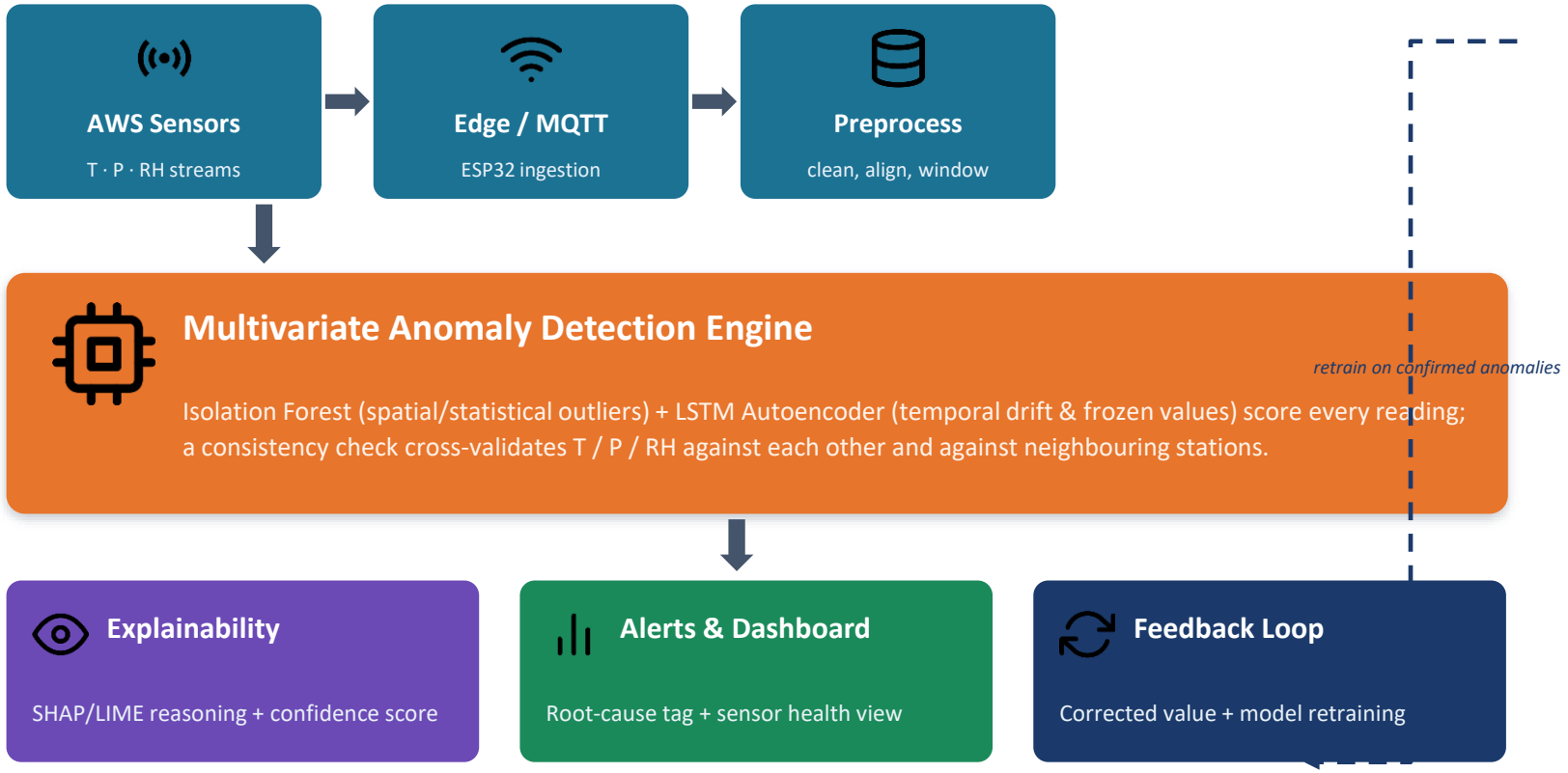
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TECHNICAL APPROACH

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TECHNOLOGIES USED

- Python, Pandas, NumPy for the data pipeline
- Isolation Forest + LSTM Autoencoder for temporal/multivariate anomaly scoring
- SHAP / LIME for explainable root-cause reasoning
- MQTT / REST for AWS → cloud streaming ingestion
- TensorFlow Lite on ESP32 for on-station edge inference
- Grafana / React dashboard for alerts & sensor health



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FEASIBILITY AND VIABILITY

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FEASIBILITY

Technical:

- Proven, lightweight stack — scikit-learn / TensorFlow Lite run comfortably on existing AWS data loggers and low-power ESP32 boards.
- Works with only 3 parameters (T, P, RH), so no new sensor hardware is required for rollout.

Operational:

- Plugs into existing AWS telemetry (MQTT/REST) without disrupting current forecasting pipelines.
- Runs online (streaming) or offline (on-station) depending on connectivity.

VIABILITY

Market & Policy:

- Directly serves IMD/MoES's AWS network and India's wider Disaster Management Framework — no new procurement needed.
- Improves data feeding NWP models, aviation, agriculture and early-warning systems alike.

Scalability:

- Same model architecture scales from a single pilot station to the full national AWS network with per-station fine-tuning.

TECHNICAL CHALLENGE

Risk: Rare/novel fault modes may not resemble any anomaly seen in training data.

Mitigation: Combine unsupervised (Isolation Forest) with semi-supervised fine-tuning, and continuously retrain on operator-confirmed alerts.

DEPLOYMENT CHALLENGE

Risk: Intermittent connectivity at remote AWS sites can delay alerts.

Mitigation: Ship an on-device edge model for offline scoring, syncing verified anomalies once connectivity resumes.

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IMPACT AND BENEFITS

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WHO BENEFITS

Disaster Management
Cleaner AWS feeds mean faster, more reliable severe-weather and flood early warnings.

Forecasting & Research
NWP models and climate researchers get fewer corrupted data points to filter manually.

Aviation & Agriculture
More trustworthy pressure/humidity readings support flight planning and irrigation decisions.

Network Operations
Sensor-health scoring flags stations needing maintenance before they fail outright.

EXISTING QC vs. SKYGUARD AI

Capability	Threshold QC	SkyGuard AI
Adapts to seasonal/diurnal patterns	⊗	✓
Cross-checks T, P & RH together	⊗	✓
Explainable, confidence-scored alerts	⊗	✓
Works on-device at the AWS station	⊗	✓
Suggests a corrected value	⊗	✓

Real-Time Streaming
Live scoring on incoming AWS telemetry

Edge on ESP32
Offline inference at low-power stations

Network-Wide View
Cross-station spatial consistency checks

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RESEARCH AND REFERENCES

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FOUNDATIONS & REFERENCES

- WMO Guide to Instruments and Methods of Observation (WMO-No. 8) — standard practice for AWS quality control and instrument siting.
- IMD Automatic Weather Station network specifications — defines the parameter ranges and reporting cadence SkyGuard AI is tuned against.
- Isolation Forest for unsupervised anomaly detection (Liu, Ting & Zhou, 2008) — basis for the statistical outlier stage.
- LSTM Autoencoders for time-series anomaly detection — basis for the temporal drift / frozen-value stage.
- SHAP — A Unified Approach to Interpreting Model Predictions (Lundberg & Lee, 2017) — basis for the explainability layer.

HOW EXISTING QC FALLS SHORT

- Fixed thresholds miss slow sensor drift and calibration decay.
- Single-parameter checks can't tell a real heatwave from a faulty thermistor.
- Most tools flag a bad reading but give no reason an operator can act on.

WHERE SKYGUARD AI STANDS OUT

- **Multivariate + spatial:** checks T/P/RH together and against neighbouring stations.
- **Explainable:** every alert carries a SHAP-based reason and confidence score.
- **Edge-ready:** same pipeline runs centrally or on an ESP32 at the station.

Project Resources: Demo Video — <add link> · Prototype — <add link> · Source Code — <add link>